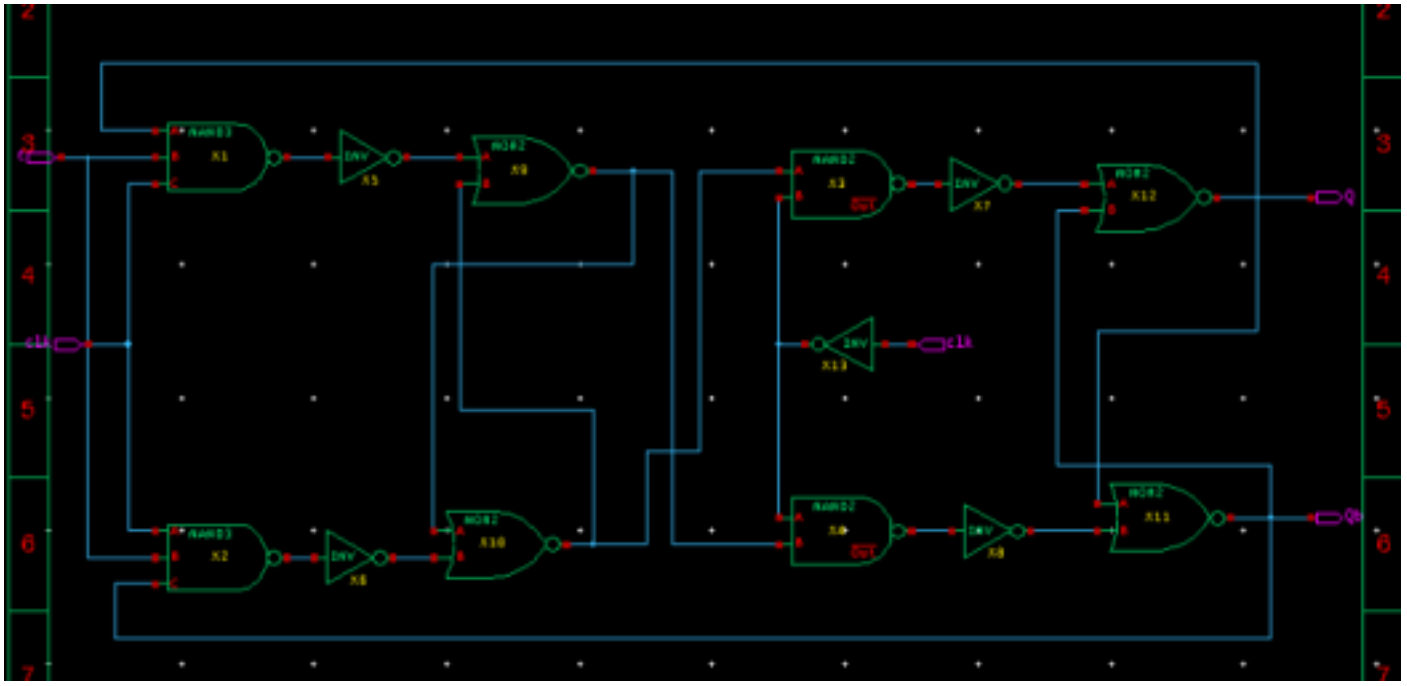
Mod-10-Counter Using T flip-flop

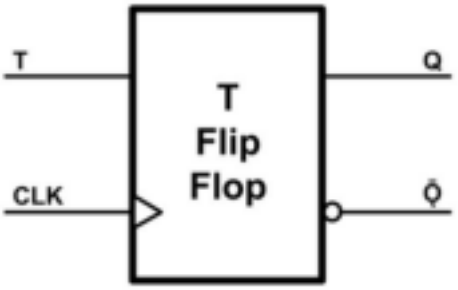
T – FLIP FLOP

• T flip flop or to be precise is known as Toggle Flip Flop because it can able to toggle its output depending upon on the input

• Toggle basically indicates that the bit will be flipped from 1 to 0 or from 0 to 1.



T	Q(n)	Q(n+1)
0	0	0
0	1	1
1	0	1
1	1	0



Usage of T Flip-Flop

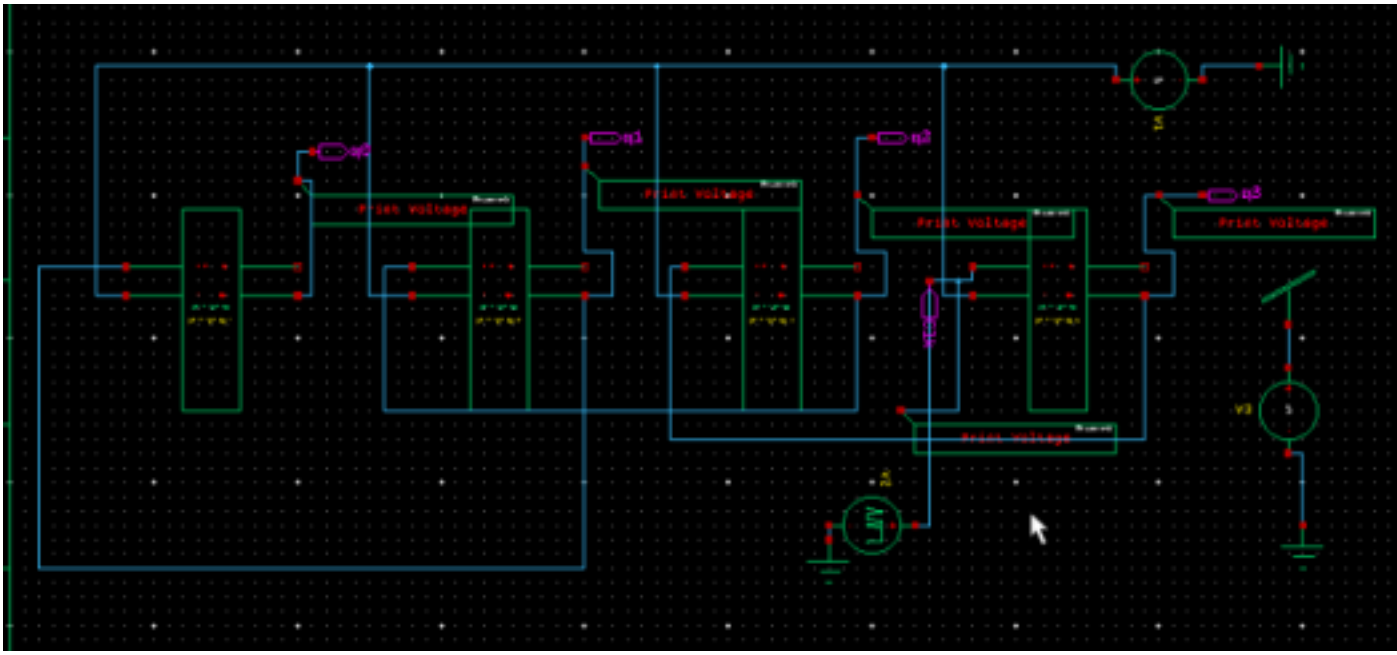
With their unique toggling feature, T flip-flops are widely used in various applications. They are a key component in digital counters, where they help to increment binary numbers. Furthermore, they are employed in digital systems for generating specific sequences and are also used in shift registers for moving binary data.

To understand the practical implementation of T flip-flops and their role in complex digital systems, it's crucial to comprehend their working principles, which is covered in the following section.

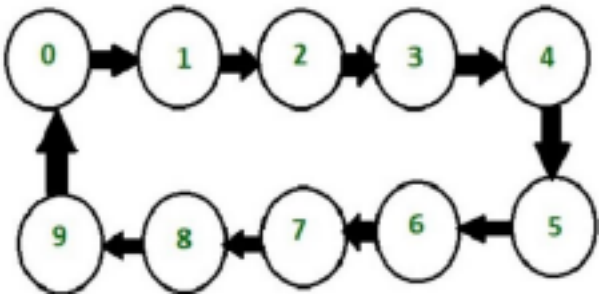
In addition to their role in counters and shift registers, T flip-flops are used in various real-world applications. For instance, they are employed in frequency dividers and in hardware design to reduce circuit complexity. Furthermore, T flip flops are also used in sequence generators and as control circuits in hardware algorithms.

MOD-10-Counter using T-flip-flop

A MOD-10 counter is a counter that goes through 10 distinct states.
It counts from decimal 0 to 9 and then resets back to 0 automatically.
It is commonly used in digital clocks and decimal counting applications.

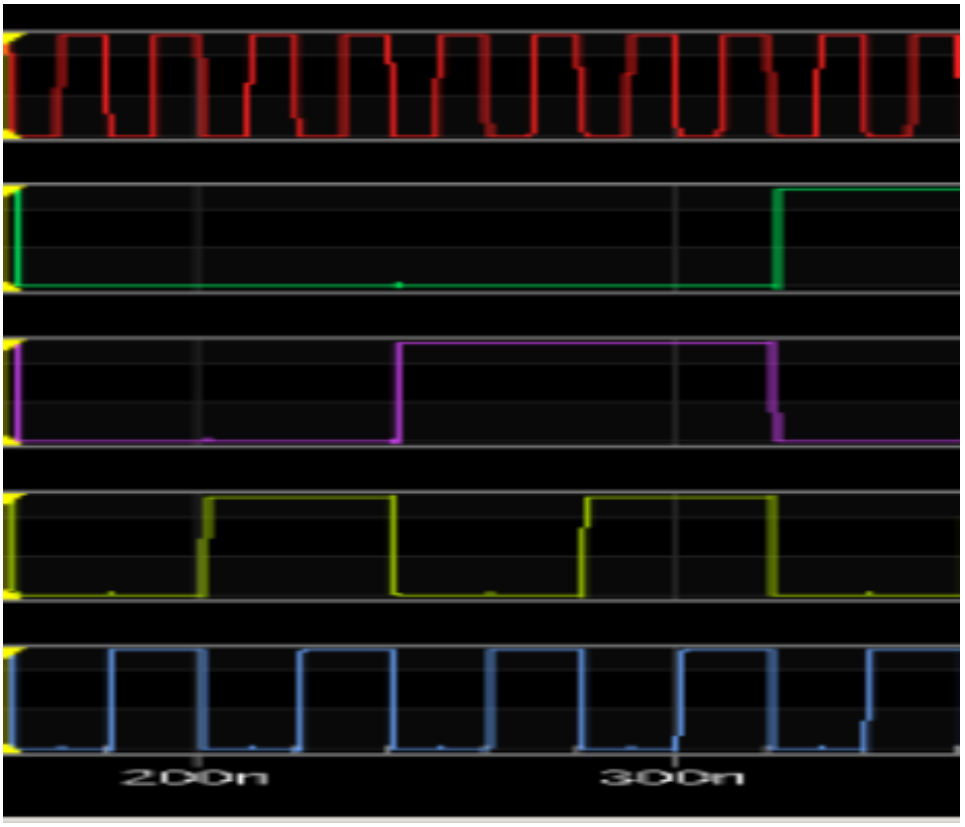


The main aim of the project is to create a synchronous counter using t – flip flops. The counter counts from 0 to 9.



Q_3	Q_2	Q_1	Q_0	Q'_3	Q'_2	Q'_1	Q'_0	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1

Output Schematic:



The MOD-10 counter using T flip-flops was successfully simulated in Tanner EDA. The output waveforms correctly counted from 0000 to 1001 and then reset to 0000. This confirms that the designed counter operates accurately and reliably.

Conclusion

The design and simulation of the MOD-10 counter using T flip-flops were successfully carried out using Tanner EDA tools. The project effectively demonstrated the toggling behavior of T flip-flops to generate a synchronous counting sequence. By incorporating suitable combinational logic, the counter was limited to count from 0 to 9 and automatically reset, achieving proper MOD-10 functionality. The simulation results confirmed the correctness and reliability of the design. This work enhanced understanding of sequential circuit design, flip-flop operation, and VLSI implementation using Tanner EDA.